Questions: Introduction to factorials, permutations, and combinations

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Summary

A selection of questions for the study guide introducing factorials, permutations, and combinations.

Before attempting these questions, it is highly recommended that you read Guide: Introduction to factorials, permutations, and combinations.

The questions below will be based on the sweet shops, Cantor's Confectionery and Lovelace's Lollies.

Q1 Factorials

1.1. In the sweet shop, Cantor's Confectionery, how many ways can you list the different types of sweets?

(a) There are 5 types of bonbons, how many ways can you list them?

(b) There are 7 types of jelly sweets, how many ways can you list them?

(c) There are 4 types of fudge, how many ways can you list them?

1.2. How many ways can you list all 16 types of these sweets?

1.3. How many ways can you list all 16 of these sweets, if you want to keep all sweets of a certain type together?

Q2 Permutations

2.1.

(a) If there are 6 types of gumdrops sold at Lovelace's Lollies, but only 3 spots on the shelf for these, how many ways could you fill these 3 spots?

(b) If there are 7 types of liquorice, but only 2 spots on the shelf for these, how many ways could you fill these 2 spots?

(c) If there are 10 types of fizzy sweets, but only 6 spots on the shelf for these, how many ways could you fill these 6 spots?

2.2. Combining all 3 of these types of sweets, how many ways could you organize the 21 different kinds of sweets into the 11 available spots?

2.3. How many ways can you do this, if you want to include at least 2 of each type?

Q3 Combinations

3.1.

(a) If there are 8 types of lollies sold at Cantor's Confectionery, how many ways could you pick 2?

(b) If there are 4 types of chocolate, how many ways could you pick 3?

(c) If there are 12 types of marshmallow, how many ways could you pick 4?

3.2. Combining all 3 of these different types of sweets, how many ways could you pick 7 sweets from the 24 different kinds?

3.3. Assuming you don't want all 7 to be marshmallows or lollies, how many ways could you pick 7 sweets from the 24 different kinds?

Q4 Counting Strategies

4.1. Lovelace's Lollies sells 3 red sweets, 3 yellow sweets, and 2 blue sweets, all different from each other. How many ways could you organize these if you don't want all 3 red sweets together?

4.2. For a display in Cantor's Confectionery, you want to arrange the 8 bonbons in a circle. How many ways are there to do this?

4.3. At Lovelace's Lollies, you want to stock 3 kinds of lollies and 4 kinds of jelly sweets. How many ways can you do this, making sure no 2 lollies are next to each other?

4.4. Say you're stocking the shelves at Cantor's Confectionery. There are 10 different types of sweets, including 5 kinds of fizzy sweets. You want to arrange these sweets on the shelf, but you want to keep all the fizzy sweets together. How many ways can you do this?

4.5. Say you're getting a bag of sweets at Lovelace's Lollies. There are 12 different types of sweets, 4 of which are fudge. You want to pick 6 sweets, but you want this to include at least 2 different types of fudge. How many ways can you do this?

After attempting the questions above, please click this link to find the answers.

Version history and licensing

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